# CHLADNI SKIN

*A Closed-Loop Haptic Nerve Feedback Device*

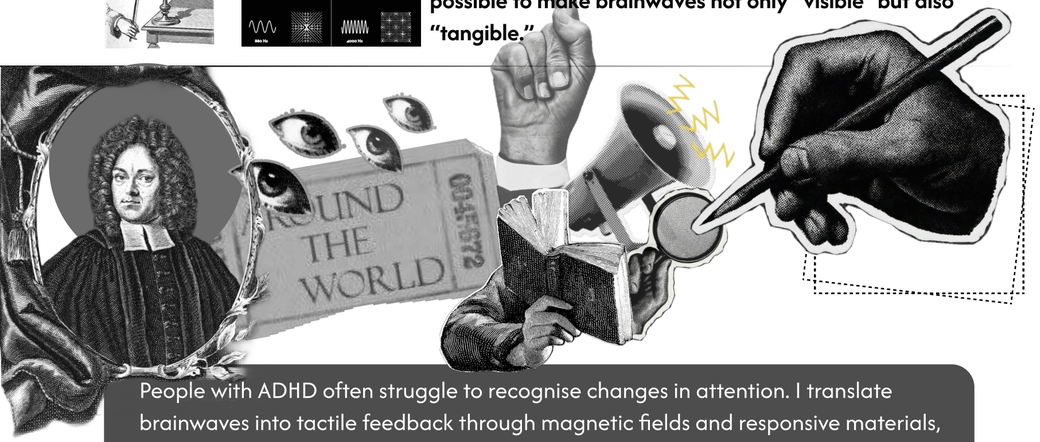
*Chladni Skin is a tactile neurofeedback device that transforms real-time EEG signals into dynamic material responses.*
*By combining soft magnetic materials, electromagnets, and haptic feedback, the project explores how invisible cognitive states can be translated into tangible touch, creating a closed-loop interaction between the brain, the body, and adaptive materials.*

RCA GRADUATE DIPLOMA - Material Culture Unit From 02/03/2026 to 27/03/2026

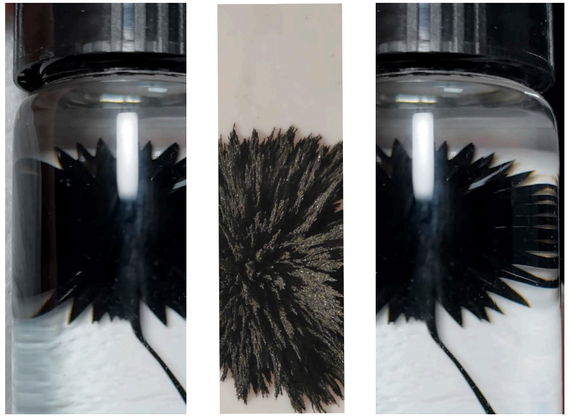
# BACKGROUND

## STARTING POINT

People have long sought to make invisible phenomena perceptible:
In 1787, Crani made sound visible through patterns; in 1846, Faraday used iron filings to visualize magnetic fields; and today, I hope to explore this further—**whether it's possible to make brainwaves not only "visible" but also "tangible."**



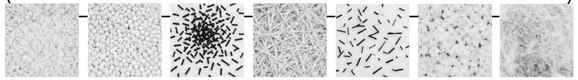

People with ADHD often struggle to recognise changes in attention. I translate brainwaves into tactile feedback through magnetic fields and responsive materials, allowing focus to be sensed through touch rather than a screen.



## MATERIAL

**A material that changes itself.**

The field does not shape the powder — it awakens it.
To translate a signal into form, I explored ferrofluid and iron powder. Under a magnetic field, they reorganize into evolving three-dimensional patterns, revealing forms already hidden within the material.



01 **Brain Signal** — Invisible shifts in attention

↓

02 **Magnetic Field** — Translate brainwaves into physical motion

↓

03 **Responsive Material** — Self-organizing transformation of iron powder and silicone

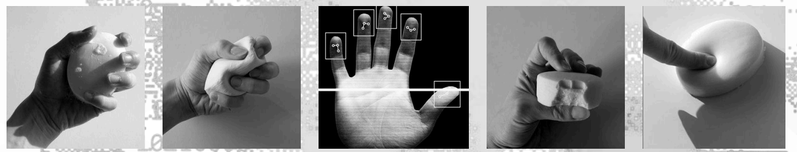
↓

04 **Haptic Feedback** — Rendering states of focus tangible and perceptible to touch

# RESEARCH

## FROM TOUCH TO FEEDBACK

- That question pulled the project out of the visual. It reminded me of the fidget toys made from non-Newtonian fluid — the "squishy" ones. As someone with ADHD, these toys are not novelties to me;they are things I actually reach for.
- The moment I recognised the material as a fidget, the project was no longer about visualising a signal. It became about making something to hold.

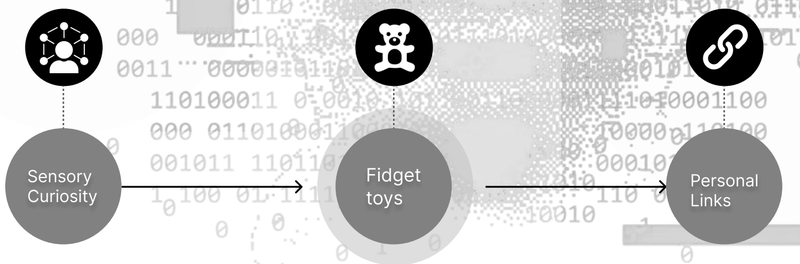


**So I started thinking:**

Is it possible to make touch itself a form of neural feedback?

## Design Opportunity

As someone with ADHD, I aim to combine the tactile experience of fidget toys with the closed-loop feedback of neurofeedback, creating a screen-free way to regulate attention through touch.




# RESEARCH METHOD

**To inform my design, I surveyed 25 adults with ADHD and compared their responses with existing fidget toys and screen-based neurofeedback devices.**

already self-regulate through touch —still carry a fidget toy, day to day.

**Can touch enable direct neural regulation**

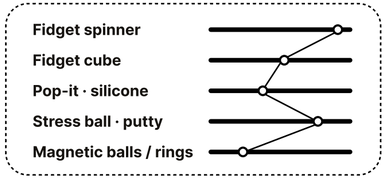
Of those who quit fidget toys: half lost the novelty, half were told it looked wrong in class or the office.wrong in class or the office.

**How to design discreet, durable tactile regulators**

have tried EEG neurofeedback. Of them,only 2 in 5 found it effective — most quit over clinic time and cost.

# RESEARCH CONCLUSION

- The research suggested that the missing link was not another device, but a responsive material capable of bridging brain activity and touch.
- Therefore, I shifted my focus from interaction design to material exploration.



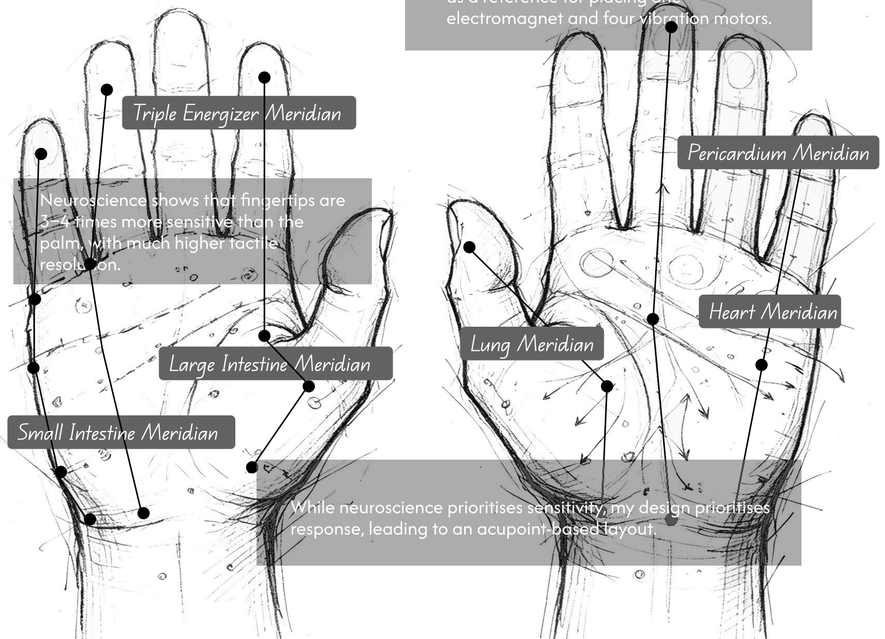

**SECONDARY RESEARCH — THE MARKET, MAPPED**

● yes ◐ partial ○ no

| | TACTILEOUTPUT | SCREEN-FREE | SOCIALLYDISCREET | SENSESYOU | CLOSED-LOOP |
|---|---|---|---|---|---|
| **Fidget spinner** | ● | ● | ○ | ○ | ○ |
| **Fidget cube** | ● | ● | ◐ | ○ | ○ |
| **Pop-it · silicone** | ● | ● | ○ | ○ | ○ |
| **Stress ball · putty** | ● | ● | ● | ○ | ○ |
| **Magnetic balls / rings** | ● | ● | ◐ | ○ | ○ |
| **Screen neurofeedback** | ○ | ○ | ○ | ● | ● |
| Chladni Skin · *working prototype* | ● | ● | ● | ● | ● |

# DEVELOPMENT

*I chose the points before*
*I understood the hand*




Traditional Chinese acupoints were used as a reference for placing one electromagnet and four vibration motors.

Neuroscience shows that fingertips are 3–4 times more sensitive than the palm, with much higher tactile resolution.

While neuroscience prioritises sensitivity, my design prioritises response, leading to an acupoint-based layout.

## Flexible Substrate Selection

After confirming the placement, I explored wearable materials and forms for a comfortable tactile experience.

*Ecoflex 00-35 Platinum Silicone*

*Silicone Stress Ball*

*Nipple Cover Silicone Film*

*Latex Condom Membrane*

*2× Medical Silicone Glove + Cycling Glove (outer structure for coil fixation and wearability)*

*2× Medical Silicone Glove · Bonded + Ferrofluid (adhesive spread uncontrollable → rejected)*

Ecoflex was selected for its flexibility, durability, fit, and stability.

## Iron Powder Carrier Exploration

dry powder

water suspension

ferrofluid

diluted ferrofluid

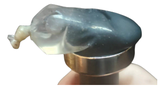
ferrofluid

diluted ferrofluid

dry ferrofluid

OIL diluted ferrofluid

dry powder



water suspension

water suspension

OIL suspension

While ferrofluid is better suited for visual effects, dry iron powder provides more reliable tactile feedback and was selected as the final material.

## Material Decision

After multiple rounds of material testing, I finally settled on a material system consisting of Ecoflex, dry iron powder, and an electromagnet.

This combination not only responds stably to brainwave-driven changes in the magnetic field but also provides tactile feedback of varying intensities while maintaining a soft texture, providing a reliable material foundation for future closed-loop neural feedback interactions.

# DESIGN

**1 Glove base fabric — substrate**

Haptic glove base with palm/wrist opening and wrist straps.

**2 Ecoflex silicone sealing membrane — outer skin**

Transparent silicone window with integrated wire channels and outer sealing slit.

**3 Iron powder layer— tactile medium, sealed inside the membrane**

Iron powder uniformly distributed make the silicone conductive.

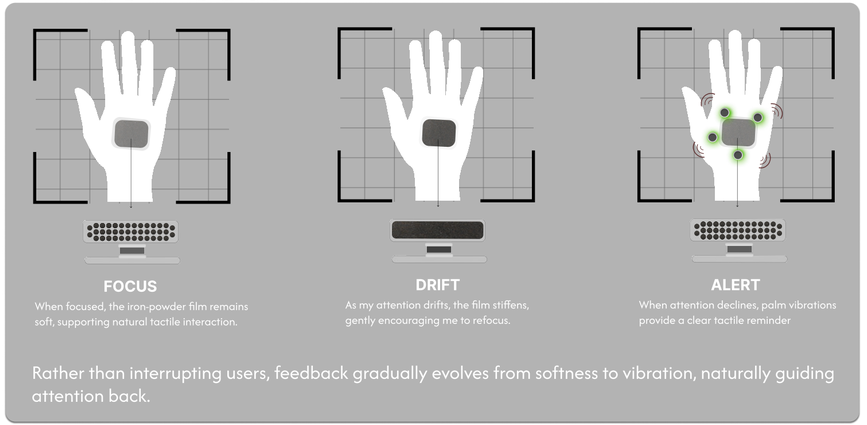
**4 Electromagnet array**

Central electromagnet coil situated in the palm center.

**5 Coin vibration motors at the four acupoint positions**

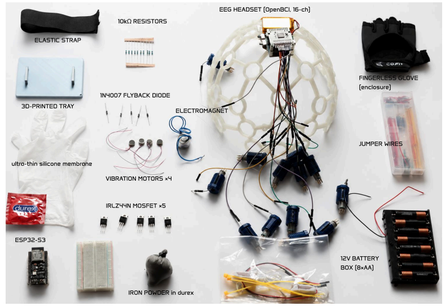
Four coin vibration motors placed around the point at the acupoint positions.

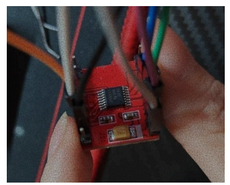
# Three Modes



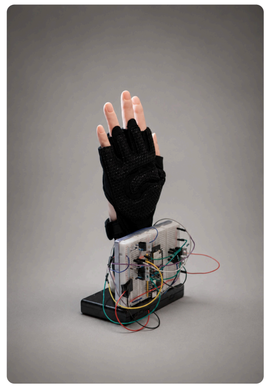

# PROCESS

```
// ── 电机引脚定义 (5V) ──
const int VM1_A = 9;   // M1 拇指 PWM 输出
const int VM2_A = 6;   // M2 食指 PWM 输出 (Shared H 的 CH7)
const int VM_HYPO = 11; // M4 小指根 (已经 M7 H (Shared H 的 CH3)
const int VM_WRIST = 10; // M4 腕部 大指 M7-7 (Board 1 的 CH7)

// 为已被占用的 CH 4(M1_A)和CH 8 (M2/D3) + 20(CH3)
const int VM_MIN = 190;
const int VM_MAX = 255;

// 模式状态
int currentMode = 0; // 0=off, 1=focus, 2=relax, 3=alert
unsigned long lastUpdate = 0;
int speedStep = 0;

void setup() {
  Serial.begin(115200);

  // 电磁铁
  pinMode(VM1_A, OUTPUT);
  pinMode(VM2_A, OUTPUT);
  pinMode(VM3_A, OUTPUT);
  digitalWrite(VM1_A, LOW);
  digitalWrite(VM2_A, LOW);
  digitalWrite(VM3_A, LOW);

  // 振动马达
  pinMode(VM_CENTER, OUTPUT);
  pinMode(VM_THENAR, OUTPUT);
  pinMode(VM_HYPO, OUTPUT);
  pinMode(VM_WRIST, OUTPUT);
  analogWrite(VM_CENTER, 0);
  analogWrite(VM_THENAR, 0);
  analogWrite(VM_HYPO, 0);
  analogWrite(VM_WRIST, 0);

  Serial.println("=== COcket Skin Control ===");
  Serial.println("Commands:");
  Serial.println("  0/1/2/3  - Mode: off/focus/relax/alert");
  Serial.println("  m(num)(val) - toggle motor 1/2/3/4");
  Serial.println("  f - FOCUS mode");
  Serial.println("  r - RELAX mode");
  Serial.println("  a - ALERT mode (3-burst test)");
  Serial.println("  s - ALL OFF");
}

void loop() {
  handleSerial();

  switch (currentMode) {
    case 1: runFocus(); break;
    case 2: runRelax(); break;
    case 3: runAlert(); break;
    default: allOff(); break;
  }
}

// ========== FOCUS mode ==========
// 20Hz 脉冲包络 (50ms周期, 25ms ON / 25ms OFF)
// 王：事心，副：大鱼际/小鱼际交替
void runFocus() {
  unsigned long now = millis();
  unsigned long phase = now % 50; // 50ms = 20Hz

  if (phase < 25) {
    // ON phase
    analogWrite(VM_CENTER, VM_MAX);
    // 每500ms切换大鱼际/小鱼际
    if ((now / 500) % 2 == 0) {
      analogWrite(VM_THENAR, VM_MAX);
      analogWrite(VM_HYPO, 0);
    } else {
      analogWrite(VM_THENAR, 0);
      analogWrite(VM_HYPO, VM_MAX);
    }
  } else {
    // OFF phase
    analogWrite(VM_CENTER, 0);
    analogWrite(VM_THENAR, 0);
    analogWrite(VM_HYPO, 0);
  }
}

// ========== RELAX mode ==========
// 10秒爆波扫描: M2(thenar) -> M1(center) -> M3(hypo) -> M4(wrist)
// 正弦渐强渐弱，配合6次/分钟呼吸
void runRelax() {
  unsigned long now = millis();
  unsigned long cycle = now % 10000; // 10s cycle
  float t = cycle / 10000.0; // 0.0 ~ 1.0

  // 每个马达在周期中占25%时间, 正弦包络
  int v1 = sineEnvelope(t, 0.0, 0.25);   // M2 thenar
  int v2 = sineEnvelope(t, 0.25, 0.50);  // M1 center
  int v3 = sineEnvelope(t, 0.50, 0.75);  // M3 hypo
  int v4 = sineEnvelope(t, 0.75, 1.0);   // M4 wrist

  analogWrite(VM_THENAR, v1);
  analogWrite(VM_CENTER, v2);
  analogWrite(VM_HYPO, v3);
  analogWrite(VM_WRIST, v4);
}

int sineEnvelope(float t, float start, float end) {
  if (t < start || t > end) return 0;
  float phase = (t - start) / (end - start); // 0~1 within window
  float val = sin(phase * PI); // 0 -> 1 -> 0
  int pwm = VM_MIN + (int)(val * (VM_MAX - VM_MIN));
  return pwm;
}

// ========== ALERT mode ==========
// 3次急促震 (100ms ON, 100ms OFF) 然后停止
void runAlert() {
  unsigned long elapsed = millis() - lastUpdate;

  if (speedStep < 3) {
    if (elapsed < 100) {
      // ON
      analogWrite(VM_CENTER, VM_MAX);
      analogWrite(VM_THENAR, VM_MAX);
      analogWrite(VM_HYPO, VM_MAX);
      analogWrite(VM_WRIST, VM_MAX);
    }
    else if (elapsed < 200) {
      // OFF
      analogWrite(VM_CENTER, 0);
      analogWrite(VM_THENAR, 0);
      analogWrite(VM_HYPO, 0);
      analogWrite(VM_WRIST, 0);
    }
    else {
      speedStep++;
      lastUpdate = millis();
    }
  }
  else {
    // 3次完成, 停止
    allOff();
    currentMode = 0;
    Serial.println("Alert burst complete");
  }
}

// ========== 全关 ==========
void allOff() {
  digitalWrite(VM1_A, LOW);
  digitalWrite(VM2_A, LOW);
  digitalWrite(VM3_A, LOW);
  analogWrite(VM_CENTER, 0);
  analogWrite(VM_THENAR, 0);
  analogWrite(VM_HYPO, 0);
  analogWrite(VM_WRIST, 0);
}
```



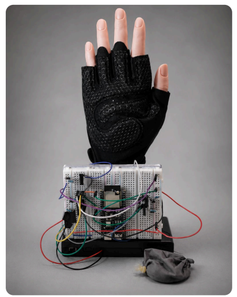



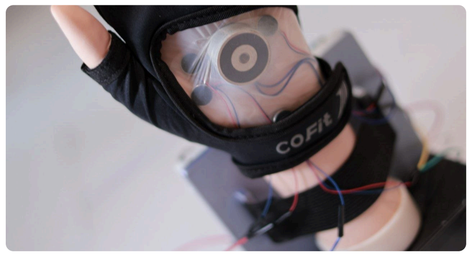
# OUTCOME

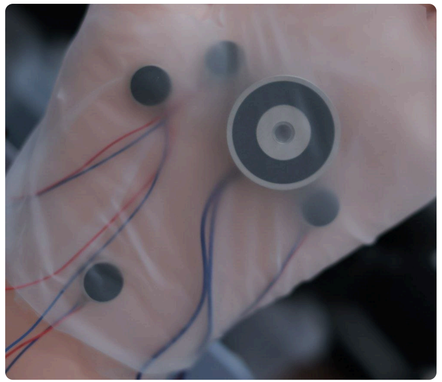


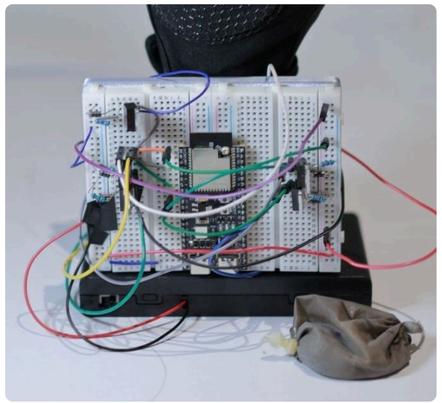
Rather than offering a solution, this project proposes another way of relating to attention—through materials, touch, and embodied interaction.

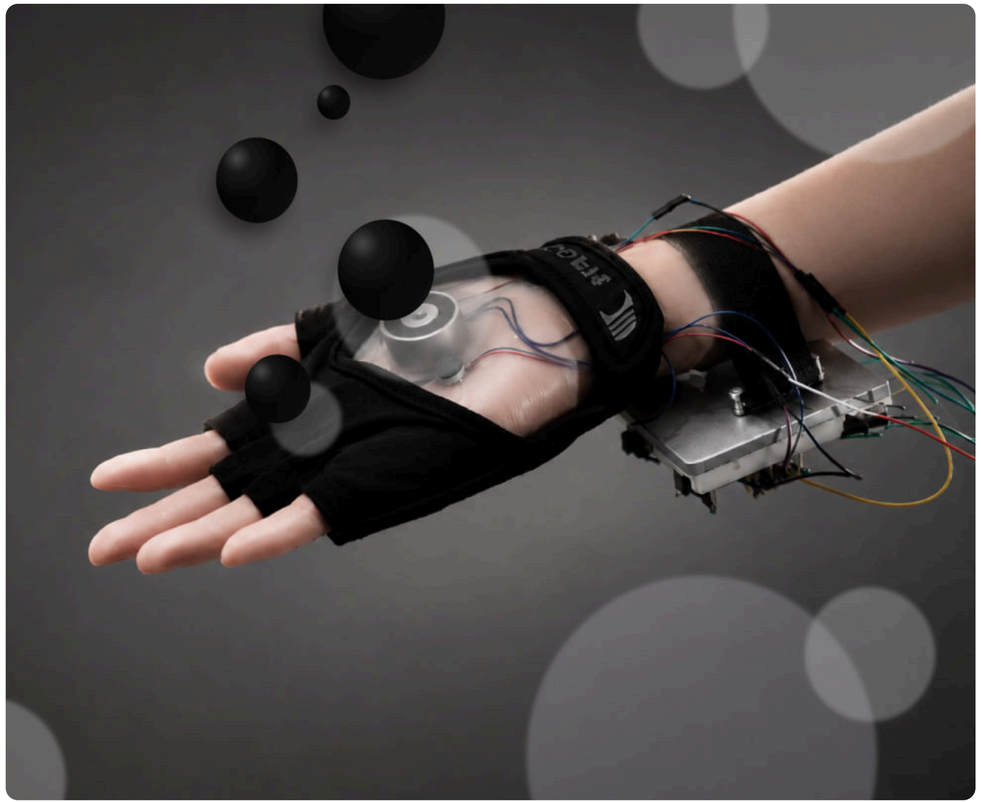
What if attention could be touched rather than measured? *Chladni Skin* is not a finished answer, but an invitation to imagine new relationships between the brain, the body, and responsive materials.











# A-LIFE

## An Ecological Environment Translation Interface

*AI Life is a physical installation that translates the health status of an ecosystem into perceptible emotional expressions for humans.*

*At its core lies a bionic egg-shaped device: it captures environmental data from a specific ecological niche (soil, freshwater, or indoor air) via physical sensors. The data is then interpreted to quantify the target species' comfort level and stress intensity through a replaceable JSON configuration file dubbed the "species genome". Finally, an LED ring renders this information into intuitive visual outputs that convey distinct emotional states.*

**RCA GRADUATE DIPLOMA – Investigate & Realise Unit  From 05/05/2026 to 29/06/2026**

# BACKGROUND

This project explores physical artificial life without digital screens, achieving ecological perception via embodied senses and time-series data. We created an egg-shaped bionic device integrated with multi-medium sensors, offline local LLM and time-series vector memory. It monitors household soil, fresh water and indoor air, converting environmental data into bio-emotional light via customizable Species Genome. Functioning as a household ecological translator, it offers mild companionship as a side effect, lowering the public's barrier to ecological observation.

**Morphology Deduction 1** | Egg-inspired Shell Outline

Smooth curved forms of oviparous organisms form the shell base to soften the rigid industrial texture of monitoring devices and endow the carrier with natural organic gentleness.

**Morphology Deduction 2** | Porous Breathable Shell Structure

Hollow shell mimics biological epidermis for air sampling and even ambient light diffusion.

**Morphology Deduction 3** | Internal Layered Sensor Compartment

Layered internal compartments accommodate detachable sensors for soil, fresh water and indoor air collection.

**Morphology Deduction 4** | Tripod Bionic Support Structure

Tripod base mimics biological limbs to stabilize the egg carrier and highlight complete artificial life embodiment.

# INSPIRATION SOURCES

## Underlying Technical References

- **Embodied Cognition Theory：** Proposed by Aristotle and Plato and refined by Descartes, it underpins the project's screen-free, body-sensing environmental perception concept.
- **Abandoning screen-based text interaction：** While conventional monitors rely on cold digital data, this project applies embodied cognition to enable the bionic carrier to sense environments through organism-like physical perception.
- **Practical Application：** It reduces users' learning cost, allowing intuitive judgment of soil, water and air conditions via gradual light changes without professional maintenance knowledge.

## Technical Logic Reference

**Technical Inspiration：**
SillyTavern Local LLM Frontend Framework

- **Interaction logic：** An anthropomorphic life system with long-term time-series memory turns ecological data into species-adaptive light rhythms via customizable Species Genome.
- **Core interaction：** No digital text; slow organic light cues visualize environmental conditions for easy user understanding.

## Visual Interaction Reference

- **Core Design Constraint：** It acts as an ecological data translator rather than an emotional companion pet, with emotional light as merely a data medium.
- **Visual Reference：** Mini organic shapes eliminate industrial coldness for home ecological monitoring scenarios.

# THEORETICAL FRAMEWORK OF ARTIFICIAL LIFE PERCEPTION

Four core theories power the egg-shaped bionic translator: somatic sensing, local computation, time-series storage, multi-medium data conversion for natural perception via light cues.

**Perception Theory | Underlying Architecture**

Physical Somatic Sensing

Digital Readouts

**Reduce learning barriers for ecological signal interpretation**

**Offline Agent · Local Brain**

Fully Offline Operation

Continuous Time-Series Computing

**Built upon the SillyTavern local LLM framework**

**Time-Series Memory System**

Time-Series Vector Database

Preserve environmental variations

**Proactively detect subtle slow ecological imbalance**

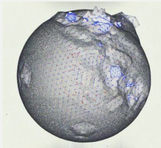
**Ecological Translation Interaction Logic**

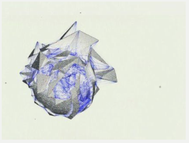
Customizable Species Genome

Translate environmental metrics

**Light & shadow: environmental information medium**

# EMOTIONAL CONNECTION

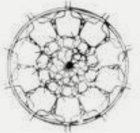




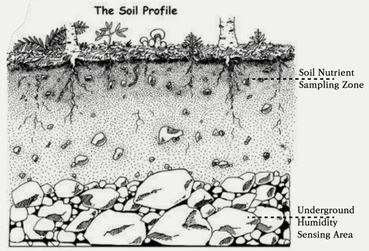
Build two-way perception channels between humans and domestic nature: multi-medium sensors capture subtle soil and water fluctuations; offline local models analyze long-term time-series environmental data; customized species genomes convert ecological comfort ranges; gradient breathing light strips visualize environmental data. It eliminates rigid digital readouts to lower ecological interpretation barriers, with companionship as an auxiliary byproduct of ecological translation.

# UNDERLYING CARRIER OF ECOLOGICAL PERCEPTION

Bionic hollow egg shell for ecological sensing. Vents for air intake & soft light, ring lights simulate creature emotion. Removable soil-water sensors inside, no screen text—environmental signals via light rhythm. Oviform curves restore life texture, realizing artificial somatic sensing.



## Soil Analysis

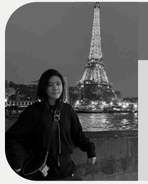



| Environmental Base Conditions | Ecological Data Computing Logic | Light & Shadow Visualization | Mood |
|---|---|---|---|
| Balanced & stable soil-water environmental indicators | All ecological values within species comfort range | 1. Slow, gentle uniform breathing gradient of light strip | ☺ |
| Unbalanced soil-water environmental indicators | Ecological values enter species stress range | 1. Irregular light rhythm for mild environmental alert | ☹ |

# FIELD RESEARCH

This field interview recruited 5 long-term plant and animal keepers. Four care relationships were categorized by bonding spectrum. Behavioral differences in perceiving ecological carriers were extracted, and typical home ecological observation behaviors summarized to support user requirements for the egg-shaped ecological interface

## ◉ Respondent 1 | Eva · Silkworm



### User Profile: Short-term Observational Caregiver

**Task-based short-term care**

**No active observation awareness**

**Only tell alive or dead**

**Observation mode：** Treat silkworms merely as curriculum research objects, follow standardized rearing procedures only; unable to spot subtle growth abnormalities or latent environmental fluctuations.

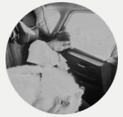
**Care deficiency：** Lacks long-term rearing experience, insensitive to subtle ecological shifts, fully dependent on rigid care routines.

## ◉ The other four stratified respondents

Has shared an eleven-year symbiotic bond with pets. Years of companionship enable recognition of subtle emotional cues, fostering profound empathy with living creatures. Desires unique, irreplicable personalized light gradients to reflect the creature's mood.

Qiqi | Summary of Research on Deep Symbiotic Caregiver

Only provides intermittent company during winter and summer vacations; can only perceive basic superficial demands such as playing and craving petting. Envisions artificial carriers as lively, free-wandering vibrant life forms.



Long-term foliage caretaker, judges environmental conditions via slow plant growth rhythm, rejects harsh flickering digital readouts, prefers soft gradual light gradients to convey ecological status.

Judges organism abnormalities based on daily routine benchmarks; detects state deviations yet fails to pinpoint environmental triggers, expecting the carrier to bridge the perceptual gap between humans and ecosystems.

Z · Samoyed | Research Summary of Intermittent Short-term Caregiver

**Eva · Silkworm**
Low emotional attachment

**Z · Samoyed**
Vacation companionship

**Ripple · Mint**
Patient interpretation

**Suwen · Cat**
Daily rhythm

**Qiqi · Dog**
Strong emotional bond

Spectrum of Connection Intensity

# PRIMARY RESEARCH

### Vision serves as humans' core channel for perceiving ecosystems.

96% of respondents visually judge the real-time status of flora, fauna and soil environments.

Based on this common perceptual habit, the carrier adopts a circular breathing light strip as its core visual information output medium.

### The public's perception of ecological changes has a time lag.

44% of survey respondents stated they only notice environmental deviations after obvious ecological changes emerge.

The carrier is equipped with a time-series memory module to continuously record subtle cumulative long-term environmental dynamics.

### First word

**Life · Companionship**
"A quiet little life" · "A friend that expresses itself silently"
"Companion object" · "Evolving companion" · "Poetic vibe"

**Machine · Function**
"State machine" · "Reminder" · "Environmental feedback node"
"Sensor pet" · "Fish tank steward" · "Automatic feedback"

**Centered · Blurred**
"Semi-living entity" · "Low-power organic vitality"
"Gradually develops its own character" · "Nonverbal bond" · "Between object and living being"

Users position the device between tool and living carrier, the core design tension of the project.

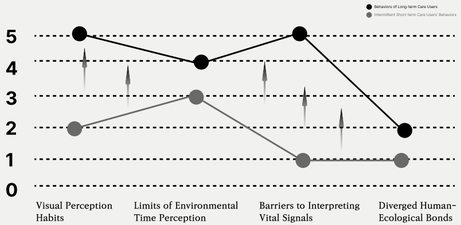
> "Caring for it reshapes my bond with the environment."

**Honest reverse signals**

20% users view the carrier as mere environment tools; two concerns become design constraints: fear of full maintenance delegation and worries over complex operation.

20% frame it as just an object; wary voices are kept as design constraints, not filtered out.

# RESEARCH SUMMARY




### Inherent Behaviors of Four Types of Users in Domestic Ecosystem Maintenance：

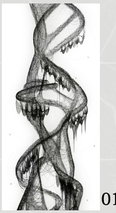
01：96% users judge environment via vision；

02：44% fail to spot subtle early environmental changes；

03：Long-term care experience is required to interpret life signals；

04：Users fall into two emotional demand tendencies toward human-ecology carriers；

**All above behavioral characteristics serve as complete design basis for the subsequent egg-shaped ecological translation interface prototype and experiments.**



01

### Root System

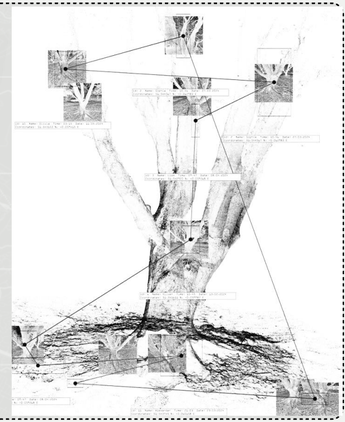
- Interviews with mint and silkworm keepers show soil and vegetation evolve slowly over long cycles. Minor imbalances build up over months, making gradual state shifts hard for ordinary people to notice consistently.
- Built-in vector time-series database stores weeks to months of continuous environmental data, recording subtle long-term ecological changes overlooked by humans.
- The ring LED strip simulates biological respiration, using soft light gradients instead of harsh numbers to reflect slow ecological growth.

### Dual Helix of Organisms

- Questionnaire data reveals 96% of users assess ecosystem conditions visually. Human sight only captures surface changes, unable to detect hidden variations like soil nutrient levels and trace water contaminants.
- To address limitations of human visual perception, this egg-shaped device integrates multiple physical sensor modules to collect underlying environmental data from soil, water and air, fully reconstructing the real ecological conditions invisible to naked eyes.

03

- Interviews with five caregivers reveal distinct nonverbal cues for pets and plants. Part-time caretakers need long-term experience to recognize subtle vital signals.
- As a standardized ecological interpreter, the device translates diverse vital signals via interchangeable species genome modules, eliminating interspecies differences and lowering learning barriers for new users.



# EXPERIMENT DESIGN

**Core question——**If an AI is given a body (sensors), continuous-time dynamics and emotional weighting, will it develop behaviour that was never explicitly programmed? And what role does socialisation / language play in it?

## 01.Direction Explored

| Direction considered | What it is | Why excluded / chosen |
|---|---|---|
| **Single-agent rearing** | Raise one agent, feed it experience, watch what it becomes | ✕ Emergence can't be attributed — can't separate embodiment, temporal continuity or socialisation |
| **Wild-only, single track** | Only the "no language, no caregiver" embodied agent | ✕ No control group; can't say whether socialisation is necessary |
| **Nurtured-only, single track** | Only the "LLM-caregiver rearing" track | ✕ No control group; can't say what a lack of socialisation does |
| **Two-track comparison (chosen)** | Wild Type × Nurtured Type run in parallel, each other's reference | ✓ The only systematic difference is socialisation / language, so any divergence can be attributed |

### Chosen Design — Two Parallel Tracks

PLAN A
**Wild Type**
Real physical sensors · no language model · no caregiver · randomly-initialised CfC network
→ What does a mind shaped by no language or society grow into on its own?

PLAN B
**Nurtured Type**
A caregiver LLM as social world + world-simulator, raising a small language model inside a structured world (a Nanjing apartment; characters Chen Ruojun, Zhou Mingzhe)
→ What does a mind raised by language and social relationships become?

## 02.Technology Selection

| Component | Options | Why excluded | Choice & rationale |
|---|---|---|---|
| **Emotional core** | ① hand-written rule state-machine<br>② Transformer<br>③ LSTM / RNN<br>④ CfC + Auto NCP | ① emotion is hard-coded "choreography", not emergent — against the premise<br>② no internal state, no time dimension, purely reactive<br>③ discrete steps, doesn't model real elapsed time | **④ CfC + Auto NCP: continuous time (dt drives the dynamics) · hidden state persists across ticks = continuity in time · small, interpretable wiring; emotion emerges from the dynamics** |
| **Memory** | ① fine-tune weights only<br>② vector DB only<br>③ off-the-shelf Mem GPT / Letta<br>④ two parallel layers | ① only statistical patterns, can't hold precise episodic memory<br>② retrievable but doesn't reshape "character"<br>③ little room to customise emotional-salience deeply | **④ Chroma episodic memory + periodic LoRA fine-tuning: mirrors human "episodic + implicit memory" — remembers what happened, and is reshaped by it into a distinct individual** |
| **Model deployment** | ① cloud API<br>② local Ollama | ① can't fine-tune / customise weight-memory · needs connectivity · privacy & persistence limited (used briefly early on as a stopgap) | **② local Ollama (Qwen3 family): fine-tunable = can be shaped by experience · offline & persistent · fully-local is a non-negotiable architectural constraint** |

**Controlled variable —** Whether the agent lives inside a socialising / language environment.

Honesty boundary — to realise "language vs no-language", the two tracks must use different architectures, so this is a comparison of two rearing conditions (closer to a feral-child thought experiment) rather than a single-variable ablation. Findings are read as qualitative / conditional contrasts, not strict causal claims.

# EXPERIMENTAL PROCESS

## Personality & Growth System — Design Decisions

### 1 · Subsystem design — what, basis, why

| Component | What was built | Basis / reference | Rationale |
|---|---|---|---|
| *5-dim attachment personality* | *secure attachment · exploration · emotion regulation · social trust · frustration tolerance* | *Bowlby / Ainsworth attachment theory + developmental psych* | *the whole question of Exp-B is whether human-like caregiving yields human-like personality; attachment theory is exactly that lens* |
| *temperament vs personality split* | *personality is seeded from innate GENOME temperament, then nudged each frame by experience via EMA* | *developmental psych: temperament (innate) vs personality (temperament + lived experience)* | *makes personality a slow experience-shaped pattern, not a momentary state; temperament fixed, personality diverges* |
| *tiered lexicon* | *282 words in 8 age-gated tiers, acquired via ambient exposure + narrative grounding* | *child language-acquisition research; vocab grows with age, learned in context* | *turns vocabulary into a smooth, measurable, comparable curve (vs the old 40-word hardcap)* |
| *language stages* | *pre-linguistic → babbling → telegraphic → simple → complex, driven by dev-progress + vocab* | *canonical child-language developmental stages* | *the granularity of psyche read-outs evolves with development, showing language being learned* |
| *psyche-behavior feedback loop* | *rule-engine psyche each frame + selective behavior; behavior feeds back into the world and next-frame perception* | *perception–consciousness–feedback loop; infant behavior repertoire* | *turns the agent from 'merely logged' into an expressive subject; per-frame LLM calls are infeasible, so a zero-cost rule engine is used* |

### 2 · Rejected alternative models

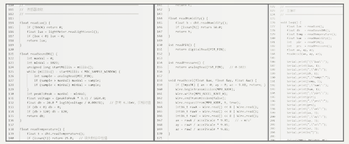
| Model | Why considered | Why rejected |
|---|---|---|
| *Big Five (OCEAN)* | *the mainstream quantitative personality model* | *adult-oriented and non-developmental; doesn't connect to caregiving/attachment, the causal lever of the study* |
| *(kept only for NPCs)* | *intuitive, handy for writing NPC personas for the LLM* | *categorical, not continuously measurable, not developmental; kept only as NPC persona flavor, never for the baby* |
| *directly tune the emotion-core weights* | *the 'cheapest' option — edit the trained CfC directly* | *would corrupt trained weights and conflate emotion (state) with personality (trait); replaced by a separate read-out layer* |
| *40-word hardcoded lexicon* | *the original version, simplest to build* | *caps instantly — produced the contradiction of '48 words at age 6 yet labeled complex sentences'; replaced by 282-word tiered acquisition* |

### 3 · Design flaws found during runs → fixes

| Flaw | Symptom & root cause | Fix |
|---|---|---|
| *world-clock desync* | *under speed-up the brain aged months while the world sat minutes back (world used real wall-clock)* | *gave the world an independent virtual clock advancing CYCLE_SEC per frame* |
| *narrative starvation* | *at 25× the local LLM couldn't supply narratives fast enough; the world idled* | *narrative backpressure (dynamic effective speed by queue) + faster playwright + pre-buffer* |
| *NPCs were mindless puppets* | *NPCs were mere props of the LLM; impossible to check if outcomes matched their persona* | *gave each NPC inner state + persona-driven appraisal + a perception–consciousness–feedback loop* |
| *connection never forms* | *caregiver presence wasn't converted into 'connection'; it stayed low → personality bottomed (root in emotion-core/separation)* | *drove the upgrade to a living multi-agent world so the low-response → low-connection chain is authentic* |
| *irreversible trait collapse* | *multiplicative decay pinned three dims near 0 with no recovery, losing resolution* | *judged a genuine persona-driven outcome; refused to reverse-engineer; soft-floor logged as future work* |
| *frozen inner monologue* | *fatigue pinned at ≈1.0 by a feedback loop (cry→tired→poor response→more cry); one repeated line* | *added crash-recovery so fatigue oscillates 0.35–0.97; reordered monologue branches, fatigue only adds a suffix* |
| *developmental-window mismatch* | *brain maxed at 2y (144k steps) yet age crept to 6, still in the nursery, burning compute* | *auto-stop at the developmental cap; reran both from birth with unified logging* |

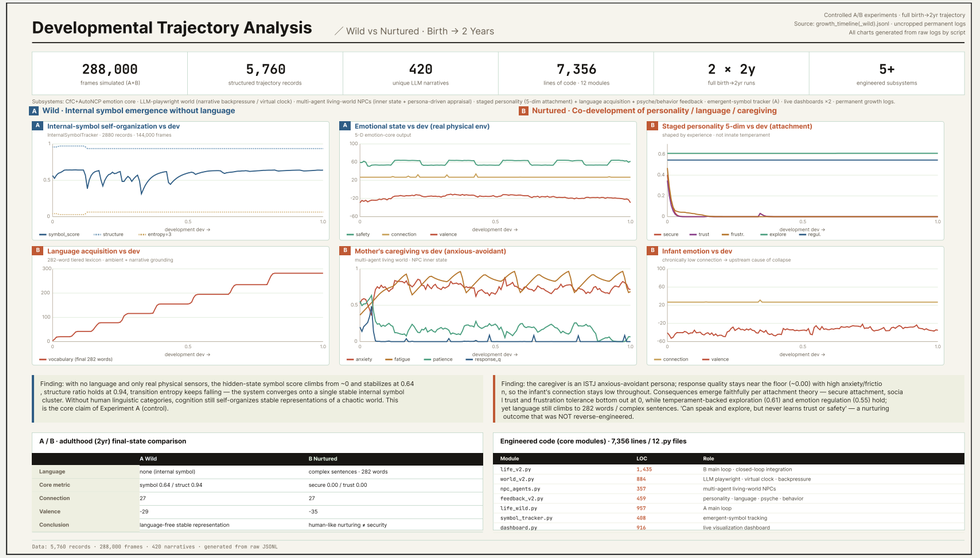
***PLAN a wild-type DASHBOARD:***

***Plana wild-type code:***

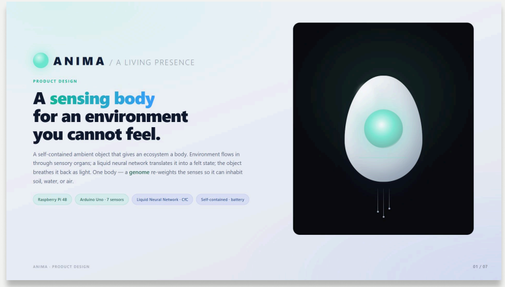


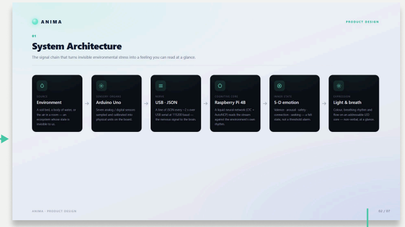
***PLAN b Nurturing dashboard:***

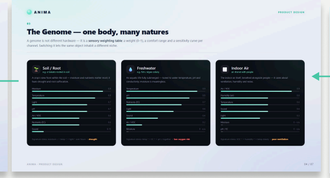
## Experimental conclusion data:

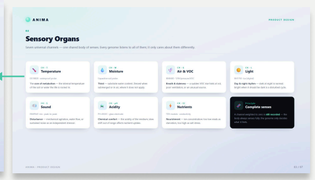


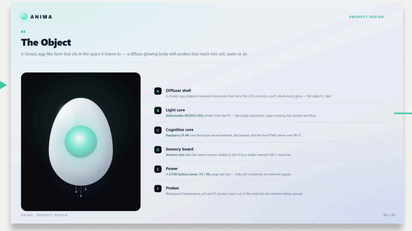
# PRODUCT INTRODUCTION

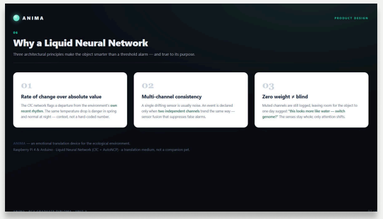





# APP DESIGN

## ANIMA——An emotional translation device for the ecological environment:

ANIMA gives an environment a body and an inner life. Streams of environmental sensor data——temperature, soil moisture, air quality, light, sound, pH, nutrients——are read on a microcontroller, weighted through a genome that defines *what this life cares about*, and translated by a liquid neural network (CfC) into a five-dimensional emotional state.

That state becomes colour, breathing rhythm, and a first-person journal of lived experience. A life is bom by choosing an ecological niche, grows continuously, can be cryo-frozen and later revived, reborn, or let go.

Mobile PWA

Real-time sensing

Raspberry Pi 4 + Arduino

Emotional interface

Liquid Neural Network· CfC

### 01 · Onboarding

Choose a genome — a life awakens and keeps it for life

Bring a life into being

### 02 · Living

How the being feels its environment, moment to moment

How it feels now

Journal & five senses

### 03 · Data

Real-time senses and the being's five-dimensional emotion

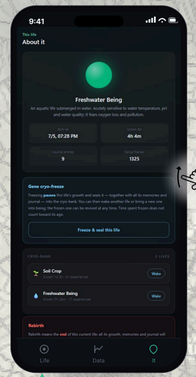
Live sensor readings

More sensors

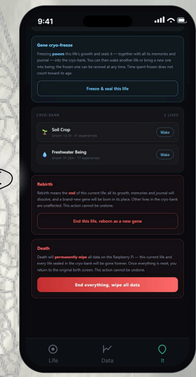
Emotion 5D & history

Identity, age, cryo-bank and life-cycle controls

### 04 · The being




Identity & cryo-freeze

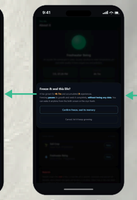



Cryo-bank · rebirth · death

### 05 · Actions

Freeze, rebirth and total reset — each guarded by confirmation

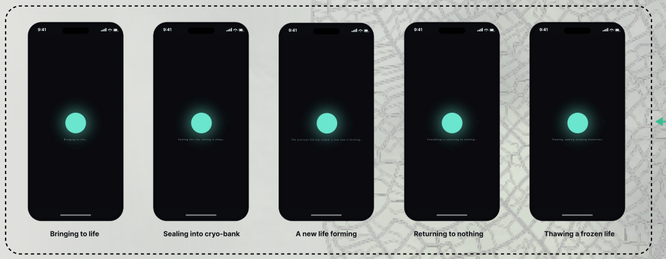
Rebirth — step 1




Freeze — confirm

### 06 · Transitions

Full-screen loading states between life events




Bringing to life

Sealing into cryo-bank

A new life forming

Returning to nothing

Thawing a frozen life

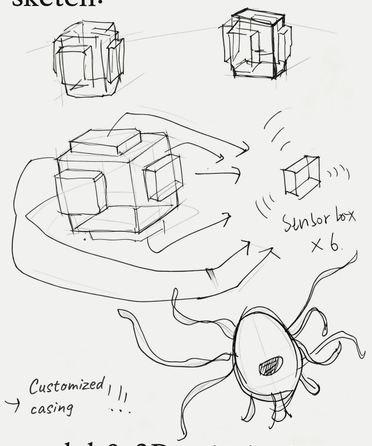
End all — final

End all — step 1

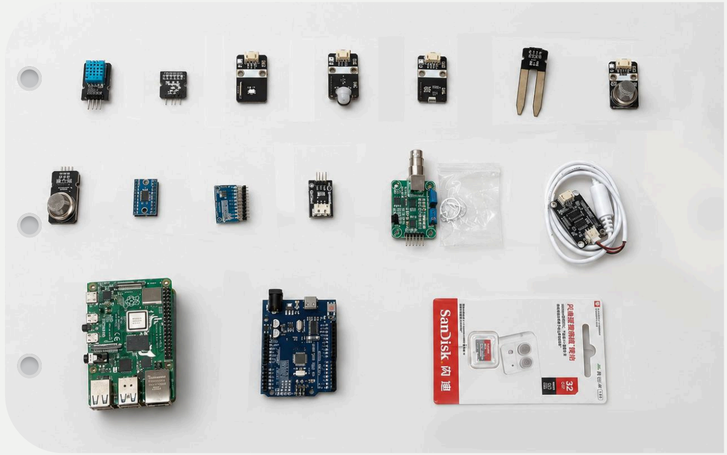
Rebirth — step 2

# PROCESS

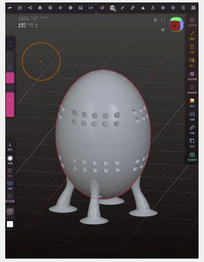
sketch:

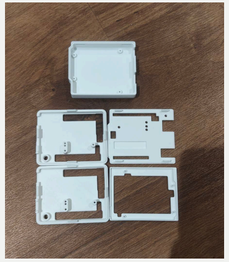



Hardware list:



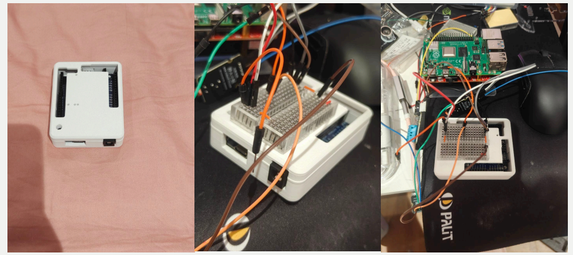
model & 3D printing:





During the process of making the shell of the arduino uno r3, I tried to modify many versions because there would be hardware sensor installation in the shell. After many modifications, the suitable size was finally determined.

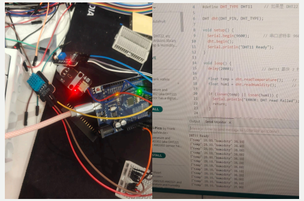
# COMPONENT ASSEMBLY

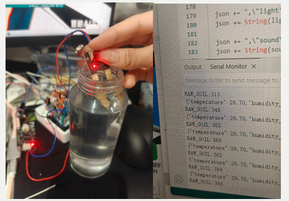


The final deliverable takes the form of a bionic egg-shaped enclosure. Its shell is 3D printed with PLA filament on a *Bambu Lab printer.* After multiple rounds of testing, the optimal scaling factor is confirmed to be between 65% and 70%.

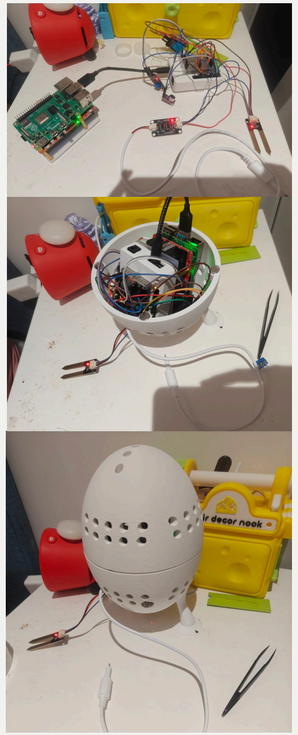
## Sensor testing:

The hardware assembly adopts a divided architecture consisting of a *computing core* and a *sensor hub.*





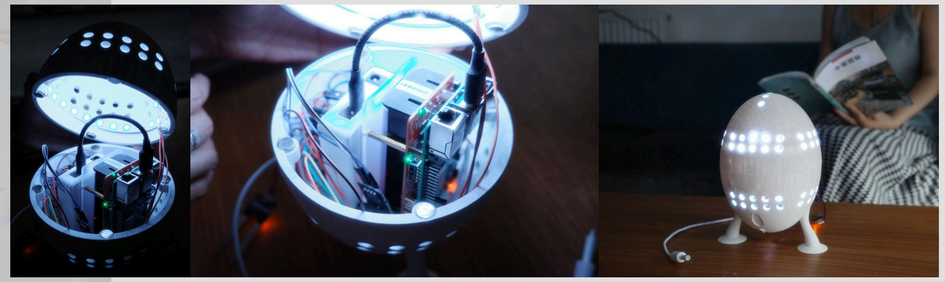
Arduino link rpi4 test:



# FINAL RESULT

This device supports switching between multiple ecological niches to collect environmental data from three categories: soil, freshwater, and indoor air.

Switching ecological niches only requires loading a different JSON file of the "species genome", with no modifications needed to the core program.



Raw sensor readings are processed via comfort curves and weight coefficients to generate a weighted deviation vector, which is then input into the emotion core. Ultimately, the LED ring translates the real-time wellbeing of the ecosystem into emotive visual feedback.

PERSONAL WORK From 1/10/2025 to 20/10/2025

# SPIRIT PATH

## A DÍA DE MUERTOS DAOIST ODYSSEY

My artistic exploration of death has never stemmed merely from a curiosity for horror or aesthetics. It was born from a profound personal and intellectual crisis of meaning—a deep anxiety triggered by mainstream modern narratives that frame death as an absolute, irreversible end.

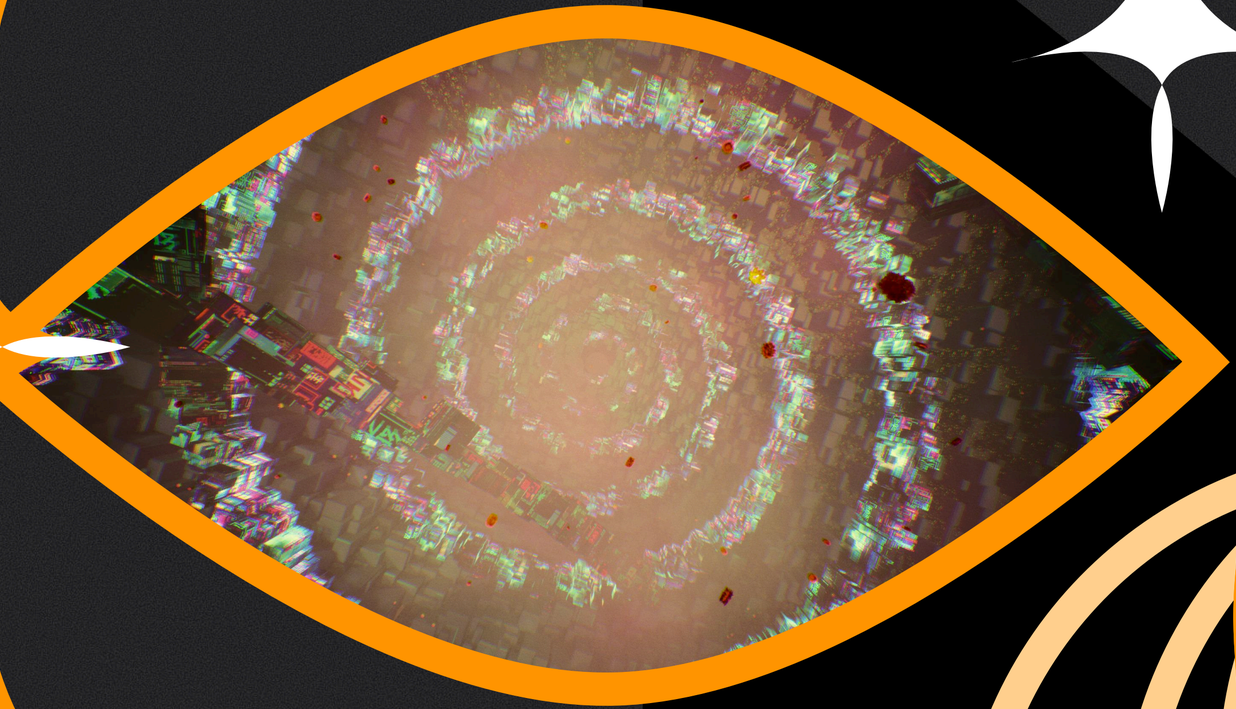
This project is my artistic attempt to construct a "third space"—a hybrid, conceptual territory blending the communal vitality of Mexican rituals with the profound void of Taoist philosophy. Through the application of UE5 and Kinect motion-sensing gameplay, I seek to materialize these abstract concepts into sensory experiences.

My work strives to respond visually and spatially to the questions posed by the mind map: Can we re-narrate this journey—not by denying death, but by celebrating life through reverence for its role in the great cycle of transformation?

Ultimately, this body of work is a personal meditation on meaning, an effort to replace existential dread of the void with a tranquil sense of belonging to a vast, interconnected whole.

https://drive.google.com/file/d/1H0I6-kXJpyKxA7tVdFcjcXPYV-rmEuCj/view

**Surrealist Media | UE Game Scene Design | Motion-Sensing Interactive Game**



# Background

## Aboo Of Death

Growing up in an environment where death was taboo, I always felt a silent anxiety about its inevitability.

**Death is a synonym for monotone**

It is a taboo subject, spoken of in whispers, and associated with loss and farewell forever.

Within such a silent perception, my encounter with Día de los Muertos felt especially profound and striking.

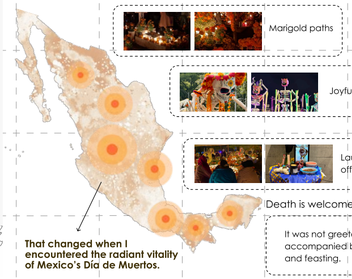
## An Encounter with Día de los Muertos

**Día de los Muertos**

Every year on November 1st and 2nd, Mexicans celebrate Día de los Muertos.
The Día de los Muertos parade is a grand celebration full of life and color.

Commemoration and Celebration

It is a culture that does not shy away from death —inviting it home with music, flowers, food, and joy.



Marigold paths

Joyful skeletons

Laughter by the offering-filled altars.

Death is welcomed with vibrant colors

It was not greeted with silence, but accompanied by music, laughter, and feasting.

**That changed when I encountered the radiant vitality of Mexico's Día de Muertos.**

## Life And Death: Cultural Similarities And Differences

**These Two Cultures—One Serene, The Other Warm And Ritualistic—Reshaped My Understanding Of Life.**

### CHINESE VIEW OF LIFE AND DEATH

**Euphemistic Language**
Talking About Death Directly Is Seen As Unlucky, So People Use Euphemisms.

**Behavioral Taboo**
Discussing Death Is Strictly Forbidden During Festivals And Celebrations.

**Emphasis on Mourning and Sorrow**
The Overall Atmosphere Is Solemn, Somber, And Mournful.

**Clear Boundaries Between Life and Death**
Memories Of The Deceased Often Reside In Isolated Corners

**Fear of Death**
Individual Death Is Seen As True Disappearance.

**Avoidant Attitude**
Death Is Generally Set Aside, With Focus Placed Entirely On Present-Life Ethics.

### MEXICAN VIEW OF LIFE AND DEATH

**HOME ALTARS**
Mexicans Believe Death Is Not The End; The Deceased Return During Día De Los Muertos To Share Joyful Nights With Their Families.

**MARIGOLD PATHS**
These Bright Orange Flowers, Called "Flowers Of The Dead," Are Believed To Guide Spirits Home With Their Scent.

**DEATH-RELATED OBJECTS**
Sugar Skulls: A Core Element Of Día De Los Muertos Rituals, Embody The Festival's Unique "Skull Aesthetic" Cultural Symbolism.

**PARADES AND FESTIVITIES**
In Mexico City, Grand Día De Los Muertos Parades Feature Participants Dressed As Skeletons In Elaborate Costumes, Dancing Joyfully To Celebrate The Continuation Of Life.

**A** INCLUSIVE MINDSET

**B** ACCEPTANCE OF THE DECEASED

**C** A RELAXED VIEW OF DEATH

**D** UNIQUE FESTIVAL CULTURE

### SHARED WISDOM OF THE TWO VIEWS ON LIFE AND DEATH

TAOIST-INSPIRED TRANSFORMATION — Life-Death Cycle
Life And Death Are Part Of A Natural Cycle.
→ Life-Death Cycle
Death Is Not The End Of Life, But A Transition To Another Stage.

TAOIST-INSPIRED TRANSFORMATION — Denial Of Ultimate Death
Death Is Not A Final End, But A Transformation.
→ Life Cycle Theory
Death Is Part Of Life; Spirits Return Annually, Completing A Reunion Cycle.

TAOIST-INSPIRED TRANSFORMATION — Ancestral Connection
Ancestors Exist In Some Form And Influence The Living.
→ Positive Life Attitude
By Celebrating Death, Life Is Glorified. Humor And Acceptance Of Death Enhance The Love For Life, Color, Food, And Family.

Maintaining Life-Death Balance Through Community Energy

**Demystifying Death**
Death is a return to the source, resting in harmony with nature.

**Serene Awareness**
The emotional tone is calm, introspective, and solemn.

**Festive Atmosphere**
An outward, joyful, and communal emotional expression.

**Positive Emotional Response**
Replace fear with gathering, sorrow with celebration, isolation with togetherness.

**Taoist View Of Life And Death**

**LIFE AND DEATH AS DAY AND NIGHT**

**DEATH AS REBIRTH**

**Mexican View Of Life And Death**

## Modern Death Anxiety

"In the post-pandemic era, death anxiety has become a widespread mental crisis."

"People both yearn to extend life through technology and feel helpless and fearful when confronted with the inevitability of death."

"The fear of death is culturally constructed."

**Why not view death from a different perspective?**

Shifts in Perceptions of Life and Death

+

How can these be transformed into an embodied, emotional experience?

Somatosensory Interaction — Immersive Gaming — Visual Design

**My project seeks to transform this understanding into an emotional and intellectual journey through an interactive experience.**

# Reserch

## History Of Día De Los Muertos

### Aztec Death Worship

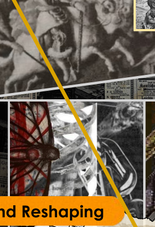
Several ancient civilizations in Central America, such as the Aztec, Maya, and Purépecha, had traditions of honoring the dead. The Aztecs worshiped the goddess of death, Mictecacihuatl, and her husband, Mictlantecuhtli, the god of the underworld.

### Forced Colonial Integration

After the Spanish invasion in the 16th century, Catholicism was imposed, labeling indigenous rituals as "pagan" and attempting to eradicate death-worship traditions.

### Mexican Indigenous Beliefs

Native cultures see death as part of the life cycle; the soul does not vanish and can connect with the living through rituals. Skulls are important religious symbols, representing the cycle of life and rebirth.

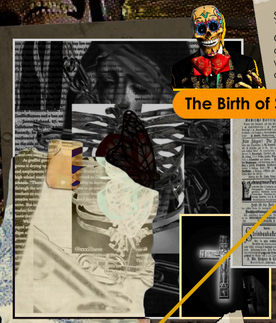


### Festival Migration and Reshaping

Catholic missionaries relocated the indigenous "Festival of the Dead" to the Catholic calendar's All Saints' Day (November 1) and All Souls' Day (November 2), aiming to overlay indigenous beliefs.

### The Birth of Sugar Skulls

Sugar skulls appeared in the 17th century, inspired by Italian Catholic funeral sweets but crafted with Mexican humor—writing names on skulls to mock death's equality.





### Cultural Resilience and Fusion

Indigenous peoples outwardly accepted the new dates but secretly preserved their core traditions.

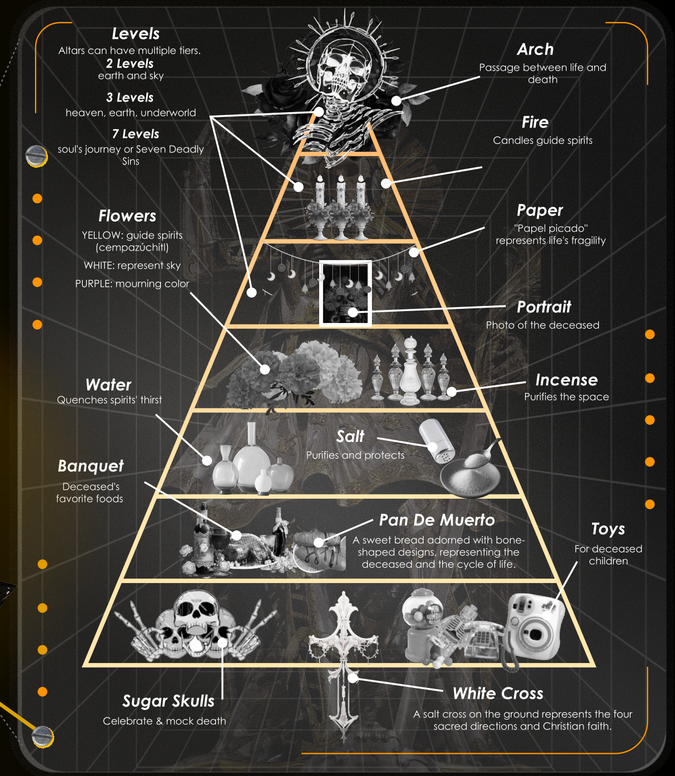
### Meaning of Modern Celebrations

Modern Día de los Muertos is a vibrant fusion of color, aroma, and memory—transcending mourning to celebrate love, life, and Mexico's unique view of death.

### Monarch Butterfly

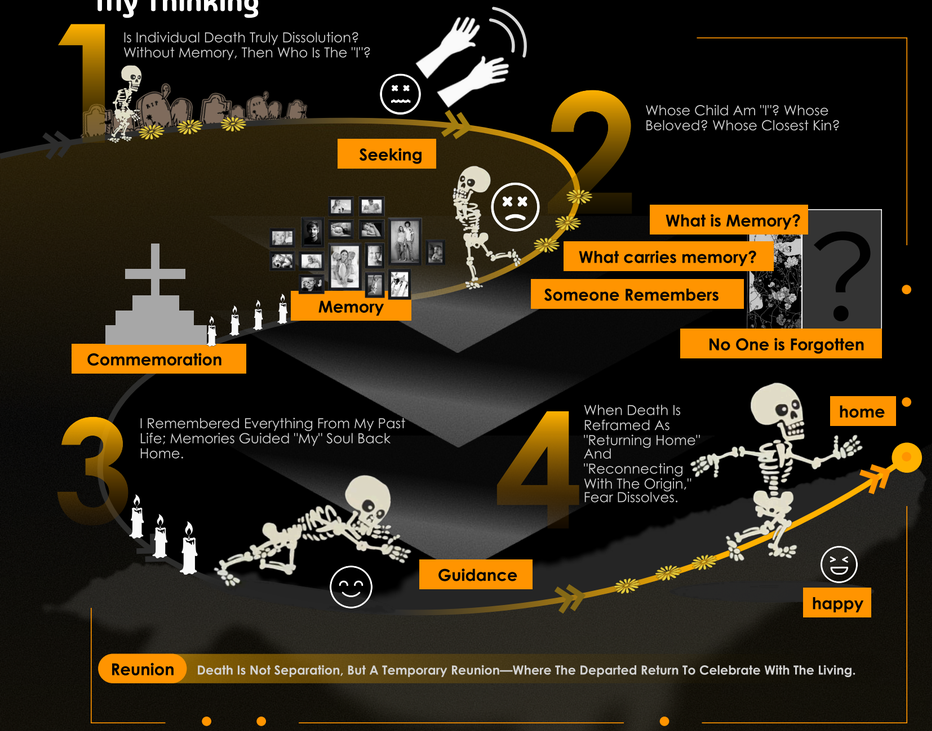
The monarch butterfly holds profound cultural significance in Mexico, where it is believed to embody the returning souls of ancestors.

## Medium For Welcoming The Departed



# Concept

## My Thinking




# The Return Path Of Souls

This Game Is My Rebuttal To The Isolation Of Death.

## Game Concept

Through Interaction, Symbolism, And Atmosphere, I Aim To Construct What Might Be Called A "Virtual Ritual"—A Digital Space That Allows Us To Confront Death Itself In A Healthier, More Connected Way.

## Game Mechanics

The Living Sustain The Existence Of The Departed Through Rituals, While The Departed Reciprocate With Blessings. This Duality Is Transformed In The Game: Living Players Solve Puzzles To Create Paths, While Departed Players Collect Memories To Achieve Transformation. The Witnessing Of The Whale's Back In The Third Level Embodies The Ultimate State Of Their Collaboration.

## Color Palette Extraction

This Image Fills My Color Palette With Vibrant Inspiration. I Have Chosen Marigold Orange As The Core Hue—Inspired By The Blossoms Covering Graves—A Warm, Saturated Tone That Defines The Festival's Visual Identity And Symbolizes Both The Light Guiding Souls And The Celebration Of Life.

Red · Marigold Pink · Marigold Yellow · Ochre · Bright Yellow · Sky Blue

Finally, I Introduced Celestial Blue And Night Black To Imbue The Integration Of The Tao With Breathability And A Spiritual Dimension, Hinting At Tranquility And Hope Beyond The Mundane.

## Symbolic Elements

Physically Deceased Individual

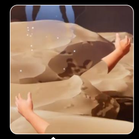
Family's Longing

The Taoist Mythical Beast Kun (Representing The Qi, Or Vital Energy, Of Human Existence)

## Game Mechanics

In the Mexican perspective on life and death, death is not the end of existence but a transition of life forms. Rooted in Aztec tradition, the departed must journey through the underworld known as Mictlán, undergoing a four-year passage to achieve rebirth.

Player 1 controls the departed soul



Player 2 controls the living

Collaborative Symbiosis

blender+UE+kinect

Life and death form a symbiosis—
the departed persist through the living's memory,
the living thrive by the departed's blessings.

The living build ofrendas(altars) offering the departed's favored food, drink, and objects—granting spiritual energy to sustain their existence.

Death is reframed as the cyclical return of life.

The departed continue to engage with the world of the living as "annual visitors".

Taoist qi-transformation teaches: Death is a return to the cosmic whole—life, decay, and rebirth in eternal cycle.

The swirling particles of qiwithin the departed echo the cyclic philosophy: "Qi gathers, life emerges; qi disperses, death returns."

## Level Instructions

The Departed Awaken

Collaborative Action

Pass Through the Passage

Return to the Mortal Realm

### Core Gameplay Logic

The protagonists—the departed and the living—must collaborate (via extra-diegetic control mechanics) to overcome challenges across all levels.

### Level 1 Instructions

Movement: Use WASD to explore the map

Puzzles: Solve mechanisms to collect items and memories

Extra-diegetic Control Trigger: When a mechanism is activated, the living player uses gesture controls to manipulate objects (by completing the puzzle's hinted solution)

### Level 2 Instructions

Flight: Extra-diegetic control activates in this level—
control the soul's left/right movement (auto-dives downward).

Collection: Touch glowing Kongming lanterns—
complete the set to pass the level.

### Level 3 Instructions

Movement: Use WASD to traverse the whale's back (no collision volume)

The final level has no combat/competition—emphasizing acceptance over conquest.

# Game Production

## Level Style

### Level 1

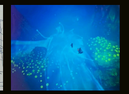
- Terrain: Marigold Sea + Altar
- Altar: Central Pentagonal Altar Aligned With Taoist Five Elements
- Calavera Skull: Iconic Symbol Of Día De Los Muertos
- Giant Candles: Guidance Mechanism
- Marigolds: Pathway For Departed Souls
- Collectibles: Memory-Bearing Objects

### Level 2

- Terrain: Whirling Vortex Passages + Architectural Silhouettes
- Destination: The Endpoint Of Return To The Mortal Realm
- Collectibles: Kongming Lanterns—Filled With The Living's Longing—Guiding The Departed's Journey Home

### Level 3

- Terrain: Vast Whale's Back + Urban Meridians
- Whale (Kun): In Taoism, The Whale (Kun) Symbolizes Transcending Limitations—
- Its "Journey To The Southern Abyss" (Zhuangzi) Metaphors The Soul's Transformative Passage.

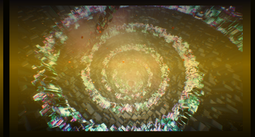


Extracting the core symbol of Día de los Muertos—the ofrenda(altar)—as an emotional foundation, the visual paradox of warm light flowing through darkness metaphorically embodies "warmth of life within death."

This level's style draws inspiration from the Temple of the Ancients in Sky: Children of the Light, reimagining the passage to the mortal realm.

Here, the whale (Kun) serves as a philosophical vessel, embodying the Taoist worldview: "All things and I are one."

Pentagonal Altar
Fuses Taoist Five Elements (wood, fire, earth, metal, water) with five return paths—a cross-cultural totem of life-death cycling.



A downward-spiraling passage, symbolizing cyclical reincarnation and rebirth.

Translucent whale → permeable life-death boundary
Whale-back perspective → living & departed coexisting in celebration

## Production Process

### Transmit Kinect-captured data to UE

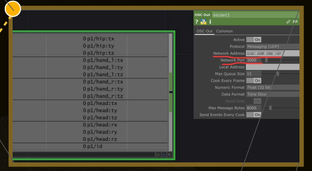
***Kinect enables the transmission of human body posture data.***

It integrates a depth sensor, RGB camera, and microphone array to capture human motion, posture, and voice in real time.

Create an OSC server in Blueprints.

Enter the IP address and port number from the TD oscout node into the corresponding fields.

Finally, create an Event Message Receiver.

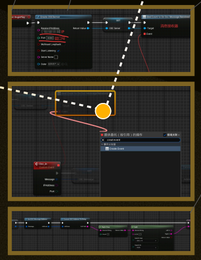


***TD Data Processing:***

1.Open the Add Operator menu → select CHOP category → create a Select operator

2.Click the arrow next to Channel Name in the select1 window

3.Choose the target skeletal bones for capture



Capture string messages transmitted via events, process them, and print the results upon completion.

***Control Components***

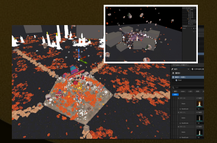
Call the created function to extract and store parameters from the X, Y, Z axes as variables.

***Implementing Kinect Interaction in UE***

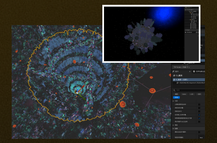
Create an Actor Blueprint, add Cube, Box, and Scene components, and position the Box component in the direction intended for the "wave-to-open" interaction.

### Model Production

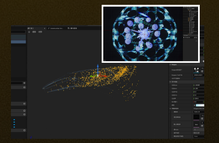
Once the game's level style is finalized, I create stylized assets in Blender.



Create a marigold sea in Blender

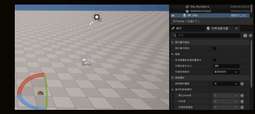


Craft spiral channels via texture mapping



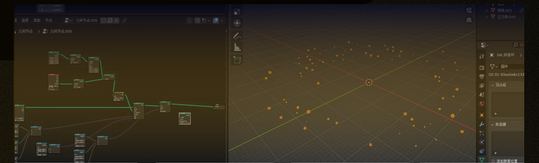
Visualize the whale (Kun) and departed souls through particle effects

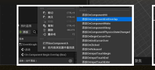
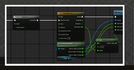
### Implement Body Tracking



Create an Actor Blueprint responsible for body tracking.

### Kinect-UE Scene Object Interaction

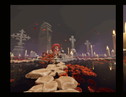


# Gaming Experience

## Level 1: Awakening

"You awaken in a sea of marigolds, your memories scattering like shattered glass.
Ahead lies a broken altar—your only guide—
holding the key to your return home."

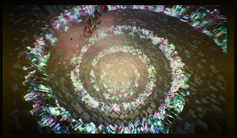


The soul, aided by the living, must solve puzzles at the central altar to activate the portal.

The living play a vital role in guiding the departed back—death is not a solitary journey, but a relationship.

## Level 2: The Return

"The spiral passage back to the mortal world descends before you,
guided by Kongming lanterns swaying in the wind—
they are the beacons of your hope ahead."



The soul descends through a passage, gathering floating Kongming lanterns—this descent leads not toward darkness, but toward life.

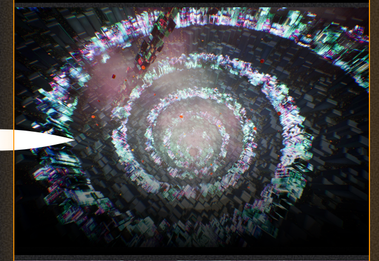
The act of collecting Kongming lanterns symbolizes converging memories and love, guiding the soul home—transforming the unknown into a passage of beauty and hope.

## Level 3: The Convergence

"You stand upon the translucent whale's back, watching the city below gradually illuminate.
No tasks, no challenges—
you need only exist, and await arrival."

The soul arrives upon the back of a vast, translucent Kun, soaring above a city celebrating Día de los Muertos.

From this serene and transcendent vantage, the soul witnesses the joy of the living below—devoid of sorrow, filled only with peace and connection.



# Spirit Path

This surreal motion-sensing game reimagines death through collaborative transcendence. Guided by Taoist qi-philosophy and the vibrant rituals of Día de los Muertos, players—both living and departed—solve environmental puzzles, gather memory fragments, and awaken mythical beings to reframe death as a cyclical return to the cosmic whole.

Built with UE5 and Kinect, the experience transforms existential anxiety into a meditative celebration of connection. Through gesture-controlled mechanics and dual-perspective storytelling, it invites players to replace dread with a tranquil sense of belonging to an interconnected whole.